



DPURandom

Random number generator.

1. Inputs:

<i>Normal</i>	= 1: The generated random numbers are normally distributed = 0: The generated random numbers have a uniform distribution. Default: 0
<i>mu</i>	If Normal = 1: mean value of the random numbers Default: 1
<i>sigma</i>	If Normal = 1: standard deviation of the random numbers Default: 0.5
<i>min</i>	If Normal = 0: smallest generated random number Default: 0
<i>max</i>	If Normal = 0: biggest generated random number Default: 1
<i>Activate</i>	The random numbers are generated if <i>Activate</i> = 1. Otherwise the DPU is not active. Default: 1

2. Parameters:

<i>Seed</i>	Seed with which the random number generator is initialized (see functional principle below). Default: -1
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3. Outputs:

<i>Out</i>	Generated random number for a simulation step.
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4. Functional principle:

The random number generator is initialized at start of the simulation with *Seed*. If *Seed* ≥ 0 the same sequence of random numbers is generated in every simulation run. If *Seed* < 0 the random number generator is initialized by the current system time.

A new random number *Out* is generated in each simulation step. If *Normal* = 0 the distribution of the random numbers is uniform within the interval $[min, max]$. If *Normal* = 1 the random numbers are normally distributed with mean value *mu* and standard deviation *sigma*.

5. Example:

```
DPUKeyboardState keyboard
{
    Computer = {LOCALHOST};
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
        Index = 1;
    };

    DPURandom random
    {
        Computer = {LOCALHOST};
        Index = 1;
        min = 1;
        max = 6;
        Seed = 42;
    };
    Connections =
    {
        keyboard.R -> random.Activate
    };
};
```

The following script generates a random number between 1 and 6 each simulation step, when „R“ is pressed on the keyboard. The random number generator is seeded with the value 42.